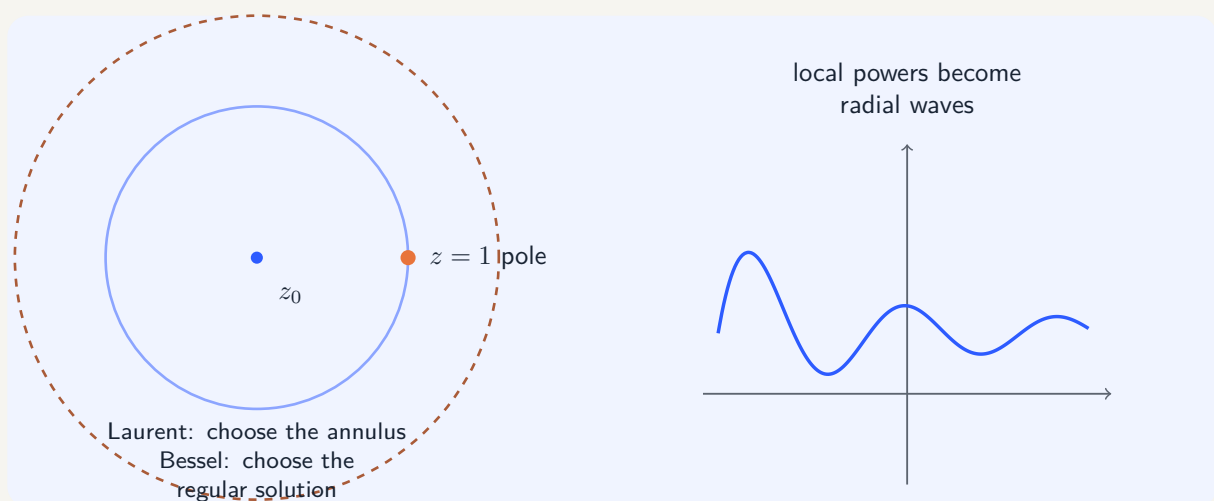


Laurent series and Bessel functions

with multivariable Taylor expansions and worked solution recipes



What this guide teaches. You will learn to (i) choose the correct Laurent annulus, (ii) derive and use the Bessel series J_ν , and (iii) build multivariable Taylor polynomials from the gradient and Hessian, with error-aware worked examples.

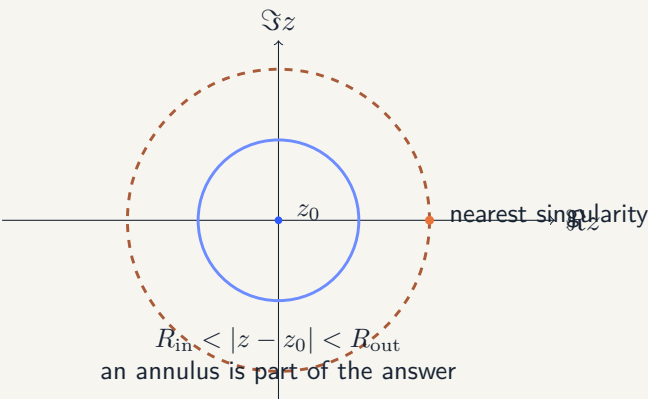
The three series are different tools

A useful way to keep the ideas straight is to ask what geometry the function is responding to.

Series	What it gives you
Taylor in one variable	A local polynomial near one point; only nonnegative powers.
Laurent in one complex variable	A local expansion on an annulus; negative powers record the singular part.
Taylor in several variables	A polynomial in the displacement vector; the Hessian records curvature and cross-effects.
Bessel J_ν	A special-function series selected by a radial differential equation.

Core discipline. Before manipulating symbols, write down the centre, the singularities, and the domain where the expansion is meant to converge. A correct algebraic identity outside its domain is not a valid series solution.

A geometric picture



Laurent series: the annulus chooses the powers

Around z_0 , a Laurent expansion has the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

The contour γ must stay inside one annulus of analyticity. The same function can therefore have different valid expansions in different annuli. **WORKED EXAMPLE A**

Expand $f(z) = \frac{1}{z(z-1)}$ about $z_0 = 0$.

- For $0 < |z| < 1$, write $f(z) = -z^{-1}(1 - z)^{-1}$.
- Insert $(1 - z)^{-1} = \sum_{n=0}^{\infty} z^n$.
- Therefore $f(z) = -z^{-1} - 1 - z - z^2 - \dots$.

The negative-power term $-z^{-1}$ is the principal part; its coefficient is the residue at 0.

For the outer annulus $|z| > 1$, factor in z^{-1} instead:

$$\frac{1}{z(z-1)} = \frac{1}{z^2} \frac{1}{1-z^{-1}} = \sum_{n=0}^{\infty} z^{-n-2}, \quad |z| > 1.$$

Check: the first two terms $z^{-2} + z^{-3}$ approximate $f(2) = 1/2$ by 0.375; adding terms moves the sum toward 0.5.

03 / BESSEL

Bessel functions: a radial ODE selects the series

The Bessel equation is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

A Frobenius trial $y = \sum_{m=0}^{\infty} a_m x^{m+r}$ gives the indicial equation $r^2 - \nu^2 = 0$. The regular branch uses $r = \nu$ and the recurrence

$$a_{m+1} = -\frac{a_m}{(m+1)(m+\nu+1)}.$$

With $a_0 = 2^{-\nu}/\Gamma(\nu+1)$, this becomes

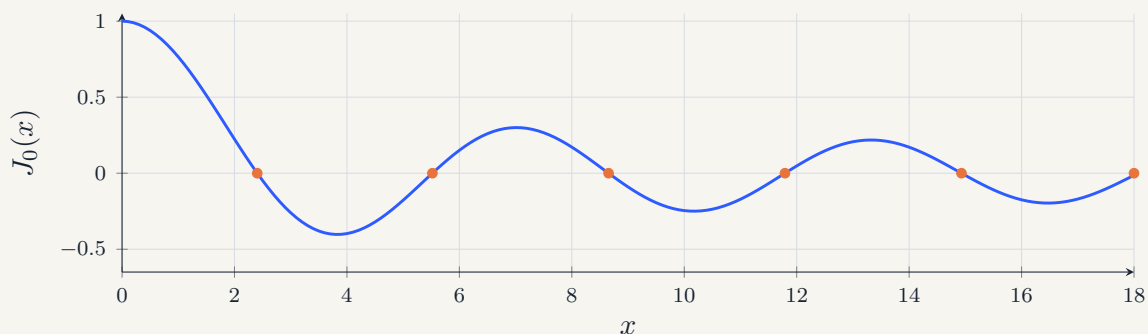
$$J_{\nu}(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m+\nu+1)} \left(\frac{x}{2}\right)^{2m+\nu}.$$

For integer ν , the second independent solution is Y_{ν} , which is singular at $x = 0$; J_{ν} is the regular choice for a finite centre. **WORKED EXAMPLE B**

Approximate $J_0(2)$ using four terms:

$$J_0(2) \approx 1 - 1 + \frac{1}{4} - \frac{1}{36} = 0.2222\dots, \quad J_0(2) = 0.2238908\dots$$

The alternating terms shrink rapidly because the ratio of successive magnitudes is $\frac{x^2}{4(m+1)^2}$ when $\nu = 0$.



04 / MULTIVARIABLE TAYLOR

The gradient and Hessian are the coefficient engine

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and expand around a . With $h = x - a$,

$$f(a+h) = f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H_f(a) h + \mathcal{O}(\|h\|^3).$$

For two variables, $h = (u, v)$ and

$$\frac{1}{2} h^T H h = \frac{1}{2} (f_{xx}u^2 + 2f_{xy}uv + f_{yy}v^2)_{(x,y)=a}.$$

The mixed derivative appears twice in matrix multiplication; that factor of 2 is a common source of errors.

Workflow. (1) choose the centre a ; (2) compute $f(a)$ and $\nabla f(a)$; (3) compute the Hessian $H_f(a)$; (4) substitute the displacement h ; (5) state the remainder order and any convergence condition inherited from the one-variable series.

WORKED EXAMPLE C

Find the second-order Taylor polynomial of $f(x, y) = e^{x+y}$ about $(0, 0)$.

$$\begin{aligned} f(0, 0) &= 1, & \nabla f(0, 0) &= (1, 1), \\ H_f(0, 0) &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \end{aligned}$$

Thus

$$T_2(x, y) = 1 + x + y + \frac{1}{2}(x^2 + 2xy + y^2) = 1 + x + y + \frac{1}{2}(x + y)^2.$$

At $(x, y) = (0.1, 0.2)$, $T_2 = 1.345$, while $e^{0.3} = 1.3498588\dots$; the error is about 4.86×10^{-3} .

[05 / MULTIVARIABLE WORKED EXAMPLES](#)

Two reusable expansion patterns

WORKED EXAMPLE D

Expand $f(x, y) = \ln(1 + x + 2y)$ through degree 2 at $(0, 0)$. Set $u = x + 2y$. Since $\ln(1 + u) = u - u^2/2 + \mathcal{O}(u^3)$,

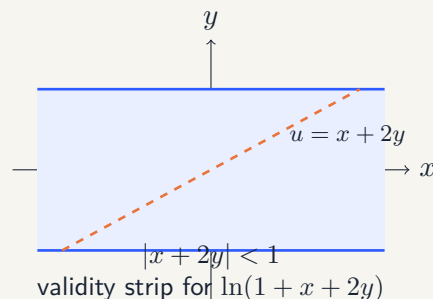
$$T_2(x, y) = x + 2y - \frac{1}{2}(x + 2y)^2 = x + 2y - \frac{1}{2}x^2 - 2xy - 2y^2.$$

The series is valid when $|x + 2y| < 1$; geometrically, this is a strip around the origin. For $(x, y) = (0.1, -0.05)$, $u = 0$ and the polynomial gives 0, exactly matching $\ln(1) = 0$. **WORKED EXAMPLE E**

Approximate $g(x, y) = \sqrt{1 + x - y}$ at $(x, y) = (0, 0)$. Use $(1 + u)^{1/2} = 1 + \frac{1}{2}u - \frac{1}{8}u^2 + \mathcal{O}(u^3)$ with $u = x - y$:

$$T_2(x, y) = 1 + \frac{1}{2}(x - y) - \frac{1}{8}(x - y)^2.$$

At $(x, y) = (0.2, 0.1)$, $u = 0.1$ and $T_2 = 1.04875$; the exact value is $\sqrt{1.1} = 1.0488088\dots$. The third-order remainder is small because $|u| \ll 1$.



[06 / SOLVE](#)

How to use the expansions to solve functions

1. **Match the geometry.** Use a Laurent series near an isolated complex singularity; use a multivariable Taylor polynomial near a regular point; use J_ν when the differential equation has circular/radial form.

2. **Turn the function into coefficients.** Factor rational functions into geometric-series pieces; differentiate the scalar prototype $\ln(1+u)$, e^u , or $(1+u)^\alpha$; or derive the Bessel recurrence from the ODE.
3. **Use the domain as a guardrail.** Record $0 < |z| < 1$, $|z| > 1$, or $|u| < 1$ next to the answer. A truncation without its domain is incomplete.
4. **Choose the truncation order from the requested accuracy.** Bound the first omitted term when the series is alternating or use a multivariable remainder estimate based on a bound for third derivatives.
5. **Validate at one point.** Compare the truncated expression with the original function numerically; this catches missing factors of 2, wrong signs, and an expansion used outside its annulus.

Radial boundary-value recipe. For $u_{rr} + r^{-1}u_r + \lambda^2 u = 0$, set $x = \lambda r$. The equation becomes Bessel's equation of order 0, so $u(r) = AJ_0(\lambda r) + BY_0(\lambda r)$. If the centre $r = 0$ is included and u must stay finite, set $B = 0$. If $u(R) = 0$, choose $\lambda R = j_{0,k}$, a zero of J_0 .

Need	First move
Pole or isolated singularity	Find the singularity distances; choose the Laurent annulus.
Local approximation in (x, y)	Compute f , gradient, Hessian at the centre.
Circular/radial oscillation	Test for Bessel's equation after a change of variables.
Accuracy estimate	Inspect the first omitted term and the domain margin.

07 / REFERENCE

Formula sheet and sources

$$\text{Laurent: } f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

$$\text{Taylor in two variables: } T_2 = f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H_f(a) h.$$

$$\text{Bessel: } x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

$$\text{Regular solution: } J_\nu(x) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \nu + 1)} (x/2)^{2m + \nu}.$$

1. NIST Digital Library of Mathematical Functions, §10.2, “Bessel and Hankel Functions,” <https://dlmf.nist.gov/10.2>.
2. NIST Digital Library of Mathematical Functions, §4.2, “Definitions and Elementary Properties” (power series and principal parts), <https://dlmf.nist.gov/4.2>.
3. J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 9th ed., McGraw-Hill, 2014. Laurent expansions and residues.
4. J. Stewart, *Calculus: Early Transcendentals*, multivariable Taylor polynomials and Hessians.

Final check. Name the centre, show the coefficients, state the convergence region, and test one numerical point. Those four lines turn a formal expansion into a usable solution.